

correlation between unilateral strength/power (e.g. one-leg hop, CMJ) and speed and effectiveness in COD tests. Training that emphasises unilateral strength and power can therefore improve agility and evasive ability. Gonzalo-Skok et al. found that an Eccentric Overload Training (EOT) programme with unilateral and multidirectional movements led to more robust COD improvements than Bilateral-Vertical training.

Reduction of inter-limb asymmetries: Differences in strength and power between dominant and non-dominant limbs are common among athletes. Excessive imbalance can increase risk of injury. Unilateral training allows each limb to be targeted specifically, helping reduce asymmetries. Cao et al. showed that Unilateral Plyometric Training (UPT) reduced strength and power asymmetries in basketball players, whereas Bilateral Training (BT) increased them.

Improvement of postural and core stability: Single-limb exercises require greater engagement of the core stabiliser muscles to maintain balance and control movement. This can lead to improvements in postural stability, which are essential for the effective execution of specific skills and for injury prevention. Dafkou et al. demonstrated that integrating eccentric, balance and core exercises into football programmes improved postural strength and stability, particularly in the non-dominant leg.

Injury prevention and rehabilitation: Unilateral training can be particularly valuable in injury prevention and rehabilitation, allowing selective strengthening of one limb or muscle group without overloading the other. Macdonald et al. found that the single-leg Roman chair hold was more effective than the Nordic hamstring curl in improving the strength-resistance of hamstrings in Gaelic footballers with previous injuries.

Considerations and potential limitations

Despite the numerous advantages of unilateral training, some considerations must be kept in mind:

Reduced absolute loads: Absolute loads in unilateral exercises are generally lower than in bilateral ones, potentially limiting the development of maximum force in global muscle groups.

Higher stability and coordination demands: Proper execution requires good stability and coordination. An adaptation period thus becomes necessary for less experienced athletes.

Probable increase in training duration: Although the total volume may be matched, performing

exercises separately on each limb can lengthen the overall duration of a training session.

Transfer to bilateral performance: While unilateral training is effective for improving unilateral performance, the transfer to bilateral actions (such as jumping or sprinting) can be less direct. Some studies (Bogdanis et al., Drouzas et al.) show bilateral gains from unilateral training, while Stern et al. observed greater improvements in the back squat following bilateral training.

Practical applications and programming

Effective integration of unilateral training into physical preparation programmes requires careful planning. Key guidelines include:

Exercise selection: Exercises should reflect the sport-specific demands and the individual needs of the athlete. It is important to include movements across different planes (sagittal, frontal, transverse) to prepare the athlete for varied stresses encountered on the field.

Volume and intensity: The volume and intensity of unilateral training should be adjusted according to objectives such as strength, power, endurance or balance. Initially, it may be useful to focus on mastering technique with moderate loads and adequate volumes.

Progression: Progression in unilateral exercises can be achieved by increasing load, reducing the support base, using unstable surfaces, or raising execution speed in plyometric drills. Progression should always be tailored to the sport, skill level and specific needs of the athlete.

Off-beat: Single-limb exercises require greater engagement of the core stabiliser muscles to maintain balance and control movement. GETTY IMAGES

